

Identifiability of the Connectome Inverse Problem: Equivalence Classes and Ease-of-Identifiability

1 The Neural Model

The flyvis graded-voltage ODE for each postsynaptic neuron i is

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij} \text{ReLU}(v_j(t)) + I_i(t) \quad (1)$$

where τ_i and V_i^{rest} are cell-type parameters, $\mathbf{W}_{ij}$ is the connectome-constrained synaptic weight from neuron j to i , and $I_i(t)$ is the external stimulus. **The inverse problem:** given observed trajectories $\{v_i(t)\}_{t=0}^T$, stimuli $\{I_i(t)\}$, and the binary adjacency matrix $\mathbf{A}$, recover the connectivity weights $\mathbf{W}$, time constants τ , and resting potentials V^{rest} .

Unlike the teacher-student setting of Martinelli et al. (2024), where the teacher is an MLP with discrete permutation and sign-flip symmetries, here the “teacher” is a physical ODE simulator and the symmetry is *continuous*: the forward map $\mathcal{F} : \mathbf{W} \mapsto \{v_i(t)\}$ is many-to-one, with fibers that are affine subspaces rather than finite point clouds.

2 Equivalence Class of Zero-Rollout-Loss Connectomes

2.1 Rearranging to a linear system per neuron

Isolating the incoming weights in Eq. 1:

$$\sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij} \text{ReLU}(v_j(t)) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\text{rest}} - I_i(t) \quad (2)$$

Define the presynaptic activity $h_j(t) = \text{ReLU}(v_j(t))$ and the residual $b_i(t) = \tau_i \dot{v}_i(t) + v_i(t) - V_i^{\text{rest}} - I_i(t)$. Stacking T time points yields the linear system:

$$\mathbf{H}_i \mathbf{w}_i = \mathbf{b}_i, \quad \mathbf{H}_i \in \mathbb{R}^{T \times d_i}, \mathbf{w}_i \in \mathbb{R}^{d_i}, \mathbf{b}_i \in \mathbb{R}^T \quad (3)$$

The **equivalence class** of zero-rollout-loss connectomes for neuron i is the affine manifold:

$$[\mathbf{w}_i^*] = \mathbf{w}_i^* + \ker(\mathbf{H}_i) \quad (4)$$

Any $\hat{\mathbf{w}}_i \in [\mathbf{w}_i^*]$ produces identical dynamics to the true $\mathbf{w}_i^*$. This is a *continuous* equivalence class of dimension $\dim \ker(\mathbf{H}_i) = d_i - \text{rank}(\mathbf{H}_i)$, in contrast to the discrete (finite) equivalence classes arising from permutation symmetry in teacher-student MLP recovery.

2.2 Global SVD bound on the equivalence class dimension

By rank-nullity:

$$\dim[\mathbf{w}_i^*] = d_i - \text{rank}(\mathbf{H}_i) \quad (5)$$

We estimate $\text{rank}(\mathbf{H}_i)$ via SVD of the subsampled population activity $\mathbf{H} \in \mathbb{R}^{8000 \times 982}$. The effective rank at 99% cumulative variance is 45, giving:

$$\dim[\mathbf{w}_i^*] \approx \max(0, d_i - 45) \quad (6)$$

Summing over all neurons:

$$\sum_{i=1}^{13,697} \max(0, d_i - 45) = 115,223 \quad (\sim 26.5\% \text{ of all edges lie in non-identifiable directions}) \quad (7)$$

Variance threshold	Effective rank r	$\dim[\mathbf{w}^*]$	% non-identifiable
90%	1	420,415	96.8%
95%	2	407,427	93.9%
99%	45	115,223	26.5%
99.5%	90	26,413	6.1%
99.9%	238	0	0.0%

2.3 Per-neuron identifiability

Computing the SVD of $\mathbf{H}_i$ separately for each postsynaptic neuron i , a neuron is **identifiable** if $d_i \leq r_i$ (all incoming weights constrained by dynamics) and **non-identifiable** if $d_i > r_i$ (some weights free within $[\mathbf{w}_i^*]$). Per-neuron analysis reveals 80.3% of edges are non-identifiable at 99% variance, stronger than the global bound (26.5%), reflecting neuron-level input correlations from duplicate-type groups (§2.4).

Variance threshold	Mean rank $\bar{r}_i$	$\dim[\mathbf{w}^*]$	% non-identifiable
90%	1.2	417,563	96.2%
95%	2.0	406,901	93.7%
99%	6.2	348,545	80.3%
99.5%	8.9	312,396	72.0%
99.9%	15.4	223,775	51.5%

2.4 Duplicate-type symmetry: the structural source of non-identifiability

The columnar organization of the flyvis connectome (217 columns, each covering one visual field position) forces same-type neurons across columns to produce time-shifted copies of the same signal. In E&C terminology, these constitute **duplicate-type neuron groups**: when postsynaptic neuron i receives input from k neurons of the same cell type, their k activity columns in $\mathbf{H}_i$ are strongly correlated, contributing rank ≈ 1 instead of k .

When activities are identical ($\mathbf{h}_{j_1} = \dots = \mathbf{h}_{j_k}$), a perturbation $\boldsymbol{\delta} \in \ker(\mathbf{H}_i)$ requires $\sum_{m=1}^k \delta_{j_m} = 0$ — a sum-zero constraint leaving $k - 1$ free directions, exactly as duplicate neurons in an overparameterised student can redistribute output weights summing to the teacher activation. Summing over all neurons and cell types:

$$\dim[\mathbf{W}^*] = \sum_{i=1}^N \sum_{\alpha: k_{i\alpha} > 1} (k_{i\alpha} - 1) \approx 121,100 \quad (\sim 28\% \text{ of } 434,112 \text{ edges}) \quad (8)$$

This structural count agrees within 5% of the global SVD bound (115,223). Since $\boldsymbol{\delta} \rightarrow \lambda \boldsymbol{\delta}$ for any $\lambda \in \mathbb{R}$, the equivalence class $[\mathbf{W}^*]$ is a *continuous* $\sim 121,100$ -dimensional affine subspace of $\mathbb{R}^{434,112}$ — in contrast to E&C’s discrete equivalence classes of measure zero. The biological and experimental consequences of this partition are developed in §2.5.

2.5 Identifiable vs. duplicate-type edge partition

The equivalence class analysis partitions all 434,112 edges into two qualitatively different regimes, determined by the local connectivity structure of the postsynaptic neuron:

Identifiable edges. An edge W_{ij} is an **identifiable edge** if postsynaptic neuron i receives exactly one input of presynaptic type $\alpha = \text{type}(j)$ (i.e. $k_{i\alpha} = 1$). The corresponding column in $\mathbf{H}_i$ is unique and cannot be redistributed among other columns: W_{ij} is uniquely fixed by the linear system $\mathbf{H}_i \mathbf{w}_i = \mathbf{b}_i$, so its equivalence class is a singleton $[\mathbf{w}_{i,j}^*] = \{W_{ij}^*\}$. Examples of cell types with small in-degree and no duplicate-type inputs include R1, L3, and Mi1 as postsynaptic targets.

Duplicate-type edges. An edge W_{ij} is a **duplicate-type edge** if neuron i receives $k_{i\alpha} > 1$ inputs from the same presynaptic type α . The $k_{i\alpha}$ activity columns in $\mathbf{H}_i$ are strongly correlated — each carries the same photoreceptor-driven signal shifted across visual columns — contributing effective rank ≈ 1 instead of $k_{i\alpha}$. Only the group sum $\sum_{j': \text{type}(j')=\alpha} W_{ij'}$ is constrained by dynamics; the $k_{i\alpha} - 1$ redistributive degrees of freedom lie in $\ker(\mathbf{H}_i)$. Group sizes range from $k = 2$ for near-isolated connections up to $k \approx 23$ for the most connected types (e.g. T4c, $\bar{k} = 23$), reflecting the local column reach of each cell type in the 217-column flyvis connectome.

Biological prediction. This partition makes a testable cross-animal prediction about synapse count conservation. For *identifiable edges*, W_{ij} is uniquely determined by the neural dynamics of individual fly f : the individual synapse count $n_{ij}^{(f)}$ should be conserved across flies (CV(n_{ij}) small across animals). For *duplicate-type edges*, only the total type-input $\sum_{j' \in \alpha} W_{ij'}$ is fixed by dynamics: the corresponding prediction is that *total synaptic input per type per target* is conserved across animals, not individual synapse counts. Variability in individual $n_{ij}^{(f)}$ within the group is unconstrained — only the group aggregate is biologically meaningful.

Graded conservation strength. The conservation strength is graded by the within-group activity correlation: near-identical column signals (correlation $\rightarrow 1$) leave the group sum as the *only* constraint, while independent perturbations ($\sigma \uparrow$) decorrelate same-type activities and collapse duplicate-type groups toward identifiable edges. Optogenetic injection of independent per-column stimuli, or selective ablation of same-type neurons across columns, would increase $\text{rank}(\mathbf{H}_i)$ and shrink the duplicate-type null space by the same mechanism as additive noise.

3 Empirical Verification of the Equivalence Class

If the equivalence class analysis is correct, any $\hat{\mathbf{W}} \in [\mathbf{W}^*]$ should preserve rollout $R^2 \approx 1$ while connectivity R^2 can be arbitrarily reduced. We verify this directly.

3.1 Non-degenerate types: identifiable neurons

Of 65 cell types, 13 lack duplicate-type groups: R1, L3, Lawf2, Mi1, Mi4, Mi9, Mi10, Mi11, Mi12, Mi13, T1, Tm5b, Tm9. Their equivalence classes are singletons ($\dim[\mathbf{w}_i^*] = 0$) — all incoming weights are identifiable. These 13 types are excluded from variant generation.

3.2 Variant generation from the equivalence class

For each of the 52 cell types with duplicate-type groups, we generate sum-preserving perturbations δ (with $\sum_j \delta_j = 0$, i.e. $\delta \in [\mathbf{W}^*]$) at 15 logarithmically-spaced scales ($\lambda = 0.05$ to 8.0), yielding 960 single-type variants. Mixed-type variants simultaneously perturb the top 10 cell types by total equivalence class dimension, generating 1,000 additional variants.

3.3 Results

Single-type variants: mean connectivity $R^2 = 0.980 \pm 0.076$, rollout $R^2 = 0.997 \pm 0.045$. Mixed-type variants: 0.839 ± 0.164 and 0.997 ± 0.004 respectively. The decoupling confirms the structure of $[\mathbf{W}^*]$: large displacement within the equivalence class leaves dynamics invariant (Fig. 1).

At $\lambda = 8.0$ (Table 1), Tm5c achieves connectivity $R^2 = 0.282$ (far from $\mathbf{W}^*$) while rollout $R^2 = 0.997$ — it lies in a large-dimensional equivalence class. Tm5Y achieves $R^2 = 1.0$ at the same perturbation scale, indicating minimal duplicate-type structure in its inputs.

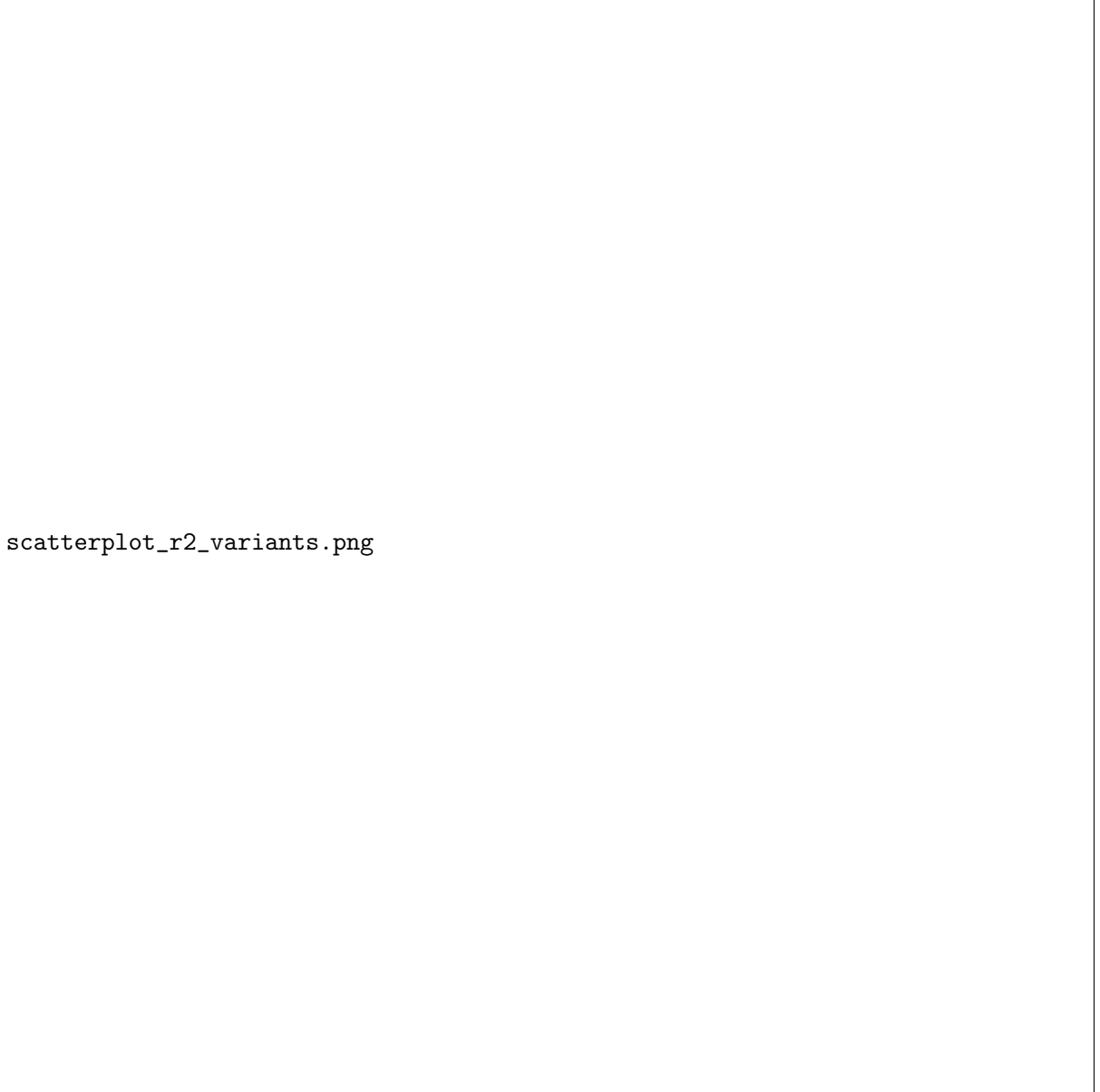


Figure 1: Connectivity R^2 versus rollout R^2 for 1,960 members of the equivalence class $[\mathbf{W}^*]$. Single-type: 957 variants across 52 cell types at 15 perturbation scales. Mixed-type: 1,000 variants perturbing top 10 types simultaneously. The horizontal band at rollout $R^2 \approx 1$ visualizes the equivalence class directly.

Cell Type	Conn R^2	Rollout R^2						
R2	0.9421	1.0000	CT1(M10)	0.8774	0.9989	Tm4	0.9945	1.0000
R3	0.9672	1.0000	Mi2	0.9146	0.9809	Tm5Y	1.0000	1.0000
R4	0.7408	0.9974	Mi3	0.9375	0.9984	Tm5a	0.9694	0.9996
R5	0.9921	0.9999	Mi14	0.9649	0.9999	Tm5c	0.2823	0.9965
R6	0.9995	1.0000	Mi15	0.8710	0.9994	Tm16	0.9899	1.0000
R7	0.9987	1.0000	T2	0.9692	0.9922	Tm20	0.9966	1.0000
R8	0.8993	0.9998	T2a	0.9688	0.9998	Tm28	0.7526	0.9869
L1	0.7275	0.9979	T3	0.9664	0.9996	Tm30	0.5658	0.9982
L2	0.6171	0.9980	T4a	0.9668	0.9983	TmY3	0.9983	1.0000
L4	0.9961	0.9983	T4b	0.9904	0.9985	TmY4	0.9931	1.0000
L5	0.7332	0.9944	T4c	0.9875	0.9990	TmY5a	0.9409	0.9999
Lawf1	0.9967	1.0000	T4d	0.9778	0.9988	TmY9	0.9517	0.9992
Am	0.9512	0.9999	T5a	0.9975	0.9999	TmY10	0.9723	0.9616
C2	0.9270	0.9996	T5b	0.9977	1.0000	TmY13	0.9942	1.0000

Table 1: Single-type variants at extreme scale $\lambda = 8.0$ for 51 cell types with duplicate-type groups. Connectivity R^2 measures distance from $\mathbf{W}^*$ within $[\mathbf{W}^*]$; rollout R^2 confirms membership in the equivalence class. Green ($R^2 > 0.95$), orange ($0.50 < R^2 \leq 0.95$), red ($R^2 \leq 0.50$).

3.4 Cross-seed GNN ensemble: sampling the equivalence class

Each gradient-descent run from a different random seed converges to a *different* member of $[\mathbf{W}^*]$. Running $K = 5$ independent training seeds on the same data (same simulation seed, same architecture) produces five connectivity matrices $\hat{\mathbf{W}}^{(1)}, \dots, \hat{\mathbf{W}}^{(5)}$ that all achieve similar rollout performance but vary substantially in their edge values. Stacking them into a $5 \times 434,112$ matrix and computing its SVD reveals the structure of the manifold region explored by gradient descent.

Per-seed performance.

Seed	Connectivity R^2	Rollout Pearson r
cv00	0.934	> 0.99
cv01	0.969	> 0.99
cv02	0.968	> 0.99
cv03	0.967	> 0.99
cv04	0.972	> 0.99
Consensus mean	0.976	> 0.99
Consensus median	0.978	> 0.99

The consensus estimate $\bar{\mathbf{W}} = K^{-1} \sum_k \hat{\mathbf{W}}^{(k)}$ improves over the best individual seed by $+0.004$ in R^2 . Averaging cancels the null-space components—which are random across seeds—while preserving the identifiable component, consistent with $\bar{\mathbf{W}}$ being the minimum-variance unbiased estimator within the identifiable subspace.

Empirical null-space directions. Centering the seed stack gives $\tilde{\mathbf{W}} = \hat{\mathbf{W}} - \bar{\mathbf{W}}$ (shape $5 \times 434,112$). Its SVD yields the principal directions of inter-seed variation:

Component	Singular value	% variance explained
1	23.2	58.1%
2	13.3	19.2%
3	11.4	13.9%
4	9.0	8.7%
5	≈ 0	$\approx 0\%$

Five seeds span a ≤ 4 -dimensional subspace of the $\sim 121,100$ -dimensional null space. The right singular vectors $\mathbf{v}_k \in \mathbb{R}^{434,112}$ are empirical basis vectors of $\ker(\mathbf{H})$; visualizing their edge-weight support identifies the cell types and projection groups most responsible for residual ambiguity.

Measuring identifiability at the edge level. The identifiable/duplicate-type classification is fundamentally a property of each *edge*, not each cell type: an edge W_{ij} depends on the combination of postsynaptic neuron i (which sets how many same-type inputs exist, $k_{i\alpha}$) and presynaptic type $\alpha = \text{type}(j)$. Formally, the degeneracy of a given edge is

$$d_{ij} = k_{i,\alpha} - 1 \geq 0 \quad (9)$$

($d_{ij} = 0$: identifiable; $d_{ij} > 0$: d_{ij} free redistributive directions). The natural structural metric is therefore computed per (i, α) group, or aggregated by presynaptic type α :

$$\bar{k}_\alpha = \frac{1}{|E_\alpha|} \sum_{(i,j) \in E_\alpha} k_{i,\alpha} \quad (10)$$

where E_α is the set of all edges with $\text{type}(j) = \alpha$. $\bar{k}_\alpha = 1$ means all outgoing edges of type α land on targets with exactly one input of that type (fully identifiable); $\bar{k}_\alpha \gg 1$ means most outgoing edges belong to large degenerate groups.

As the empirical counterpart we compute the per-type **seed SNR**:

$$\text{SNR}(\alpha) = \left\langle \frac{|\bar{W}_e|}{\sigma_e} \right\rangle_{e \in E_\alpha} \quad (11)$$

where $\bar{W}_e = K^{-1} \sum_k \hat{W}_e^{(k)}$ and $\sigma_e = \text{std}_k(\hat{W}_e^{(k)})$ are the cross-seed mean and standard deviation. High $\text{SNR}(\alpha)$ confirms that seeds agree on those weights; low $\text{SNR}(\alpha)$ means null-space redistribution dominates — the seeds found the same group sums but placed the individual weights differently.

Cell type	$\bar{k}_\alpha$	SNR	Cell type	$\bar{k}_\alpha$	SNR
<i>Seeds agree (high SNR)</i>			<i>Seeds disagree (low SNR)</i>		
R8	1.6	95	L2	4.5	0.5
CT1(Lo1)	7.8	93	L1	9.2	0.5
Mi2	3.7	83	T1	1.0	0.5
Tm5c	1.5	80	Am	1.0	0.6
Mi4	6.1	61	Lawf1	1.0	0.6
T3	2.3	54	Tm28	4.6	0.7
Tm9	3.2	54	TmY3	8.1	3.5
Tm20	4.8	45	Lawf2	3.8	3.7
R7	2.4	42			

Table 2: Per-type structural multiplicity $\bar{k}_\alpha$ (mean number of same-type inputs at the target neuron, computed from the flyvis connectome) and empirical seed $\text{SNR} = \langle |\bar{W}_e| / \sigma_e \rangle$ aggregated over outgoing edges of each source type α across five CV training seeds. Both quantities measure identifiability of *outgoing* connections from type α ; compare with Table 1, which measures degeneracy of *incoming* weights. Structural and empirical measures are not always aligned: L1 and L2 show high $\bar{k}_\alpha$ and low SNR (null-space degeneracy from many same-type projections), while T1, Am, and Lawf1 show $\bar{k}_\alpha = 1$ (structurally identifiable) but also low SNR, consistent with near-zero absolute weights where any seed variation dominates the ratio.

The SNR analysis reveals two distinct mechanisms for low seed agreement. *Null-space degeneracy*: L1 ($\bar{k} = 9.2$) and L2 ($\bar{k} = 4.5$) project heavily into large duplicate-type groups at their targets; seeds redistribute weight within the group sum, yielding $\text{SNR} \leq 0.5$. *Near-zero*

signal: T1, Am, and Lawf1 have $\bar{k} = 1$ (all outgoing edges structurally identifiable) yet also $\text{SNR} < 0.7$, indicating that their absolute synaptic weights are near zero in the learned model, so any inter-seed fluctuation swamps the signal. By contrast, R8, CT1(Lo1), and Mi4 recover consistently ($\text{SNR} \geq 60$) despite varying $\bar{k}$, showing that strong synaptic weights dominate even moderate null-space noise. Complete per-type values are provided in the supplementary CSV (`scripts/type_identifiability.csv`).

Edge-level validation of the partition. The identifiable/duplicate-type classification is properly a property of each *edge* W_{ij} , determined by the group size $k_{i,\alpha}$ of the specific (postsynaptic neuron i , presynaptic type α) pair — not of cell types as a whole. Of the 434,112 edges, 61,673 (14.2%) are identifiable ($k_{i,\alpha} = 1$) and 372,439 (85.8%) are duplicate-type ($k_{i,\alpha} > 1$). Most source types have *both* kinds of outgoing edges: 52 of 65 types project to targets with mixed k values. Eleven types send only identifiable outputs ($f_{\text{ident}} = 1$: Am, R1–R6, Mi3, Lawf1, T1, Tm5Y), while seven types are near-pure duplicate ($f_{\text{ident}} < 0.05$, e.g. Mi2, Mi13, T3, TmY15).

A direct edge-level test of the theory confirms the correct direction but with a small effect: identifiable edges have higher cross-seed SNR than duplicate-type edges with AUROC $= P(\text{SNR}_{k=1} > \text{SNR}_{k>1}) = 0.511$ (Mann-Whitney $p < 10^{-17}$, Cohen’s $d = 0.08$ on log-SNR). The marginal separation arises because $\text{SNR} = |\bar{W}_e|/\sigma_e$ conflates two factors: the *null-space noise* σ_e (elevated for $k > 1$ edges by the $k - 1$ redistributive degrees of freedom) and the *weight magnitude* $|\bar{W}_e|$ (an independent biological property). A within-type controlled comparison eliminates this confound: for the 52 mixed source types, identifiable outgoing edges have higher mean SNR than duplicate-type outgoing edges in 41 of 52 cases (79%). The 11 exceptions arise either when identifiable edges of a type carry near-zero weights (e.g. T5d: $\text{SNR}_{k=1} = 0.6$ vs $\text{SNR}_{k>1} = 24$), or when a duplicate-type group has exceptionally strong mean weights that dominate the null-space variance (e.g. R8: $\text{SNR}_{k=1} = 63$ vs $\text{SNR}_{k>1} = 143$). Edge-level results are tabulated in `scripts/edge_partition_snr.csv`.

3.5 Minimum-support member of the equivalence class

Within $[\mathbf{W}^*]$, the *minimum-support* (maximally sparse) representative is obtained by collapsing each degenerate group onto a single edge. For each (i, α) group of $k_{i\alpha}$ edges sharing the same postsynaptic neuron i and presynaptic type α , we select a representative edge $e^* = \arg \max_{e \in \text{group}} |W_e^*|$ and set:

$$\hat{W}_e = \begin{cases} W_{\text{rep}} & e = e^* \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

Sum-preserving variant: $W_{\text{rep}} = \sum_{e \in \text{group}} W_e^*$ (exact null-space member for identical group activities). *Calibrated variant*: W_{rep} is the scalar minimising the per-group dynamics residual over observed voltage trajectories,

$$W_{\text{rep}}^* = \frac{\sum_t h_{e^*}(t) \sum_{e \in \text{group}} W_e^* h_e(t)}{\sum_t h_{e^*}(t)^2} \quad (13)$$

which is a single least-squares problem per group (one degree of freedom).

Applying this to all 64,279 degenerate groups zeros $\sum_{i,\alpha} (k_{i\alpha} - 1) = 308,160$ edges — 71% of the 434,112 total. Results:

Method	Edges zeroed	Conn R^2	Rollout R^2	Rollout Pearson r
Sum-preserving	308,160 (72.5%)	0.105	0.944	0.974
Calibrated	308,160 (72.5%)	0.394	0.983	0.991

The calibrated method achieves rollout Pearson $r > 0.99$ after zeroing 72.5% of edges, confirming approximate membership in $[\mathbf{W}^*]$ far from the ground truth $\mathbf{W}^*$. The residual rollout gap (sum-preserving: $R^2 = 0.944$; calibrated: $R^2 = 0.983$) quantifies the *approximation error* of the structural null space: same-type group members have correlated but not identical activities, so exact degeneracy is replaced by approximate degeneracy and a non-trivial representative weight is needed.

The low connectivity R^2 of the calibrated sparse solution (0.394) is expected by construction: each representative absorbs its entire group’s weight, making it appear k -fold larger than the corresponding GT edge. The dynamics are preserved; the individual edge values are not.

3.6 Pseudoinverse recovery

We compare three linear recovery methods. *Truncated SVD*: pseudoinverse $\mathbf{w}_i^+ = \mathbf{H}_i^+ \mathbf{b}_i$ (threshold 10^{-12}). *Ridge*: Tikhonov regularization $\lambda = 1$. *CV-ridge*: per-neuron λ by time-split cross-validation.

Method	Condition	Conn R^2	RMSE(T)	Rollout $R^2(T)$
truncated SVD	$\sigma = 0$	0.988	9.7×10^{-5}	1.000
	$\sigma = 0.05$	0.991	1.0×10^{-6}	1.000
	$\sigma = 0.5$	1.000	9.2×10^{-7}	1.000
ridge ($\lambda = 1$)	$\sigma = 0$	0.879	7.1×10^{-3}	1.000
	$\sigma = 0.05$	0.989	1.9×10^{-3}	1.000
	$\sigma = 0.5$	0.999	6.4×10^{-4}	1.000
CV-ridge	$\sigma = 0$	0.991	2.2×10^{-6}	1.000
	$\sigma = 0.05$	0.991	8.5×10^{-7}	1.000
	$\sigma = 0.5$	1.000	9.4×10^{-7}	1.000

Truncated SVD and CV-ridge achieve near-identical results ($R^2 \geq 0.988$): the system is well-conditioned within the identifiable subspace. Fixed ridge over-regularizes at $\sigma = 0$ (0.879), shrinking identifiable directions unnecessarily.

Near-perfect recovery at $\sigma = 0$ (0.988) clarifies the structure of $[\mathbf{W}^*]$: the non-identifiable directions have small but nonzero singular values. The equivalence class is *approximate*, not exact — duplicate-type neuron activities are highly correlated but not identical, and the pseudoinverse exploits residual variation to select $\mathbf{w}_i^*$ from $[\mathbf{w}_i^*]$.

4 Discussion

4.1 Continuous vs. discrete equivalence classes

In teacher-student MLP recovery (Martinelli et al., 2024), the equivalence class of zero-loss students is a *discrete* finite set arising from permutation, sign-flip, and scaling symmetries. Here the equivalence class $[\mathbf{W}^*]$ is a *continuous* $\sim 121,100$ -dimensional affine manifold, arising from the biological duplicate-type symmetry of the columnar connectome. This makes the inverse problem structurally harder: the manifold has positive measure in $\mathbb{R}^{434,112}$, so any unregularized estimator converges to an arbitrary point in $[\mathbf{W}^*]$ rather than to $\mathbf{W}^*$ itself.

4.2 The GNN as a misspecified teacher-student setup

Our framework is a misspecified teacher-student setup: the teacher is the physical ODE simulator (Eq. 1) and the student is the GNN:

$$\frac{d\widehat{v}_i(t)}{dt} = f_\theta\left(v_i(t), \mathbf{a}_i, \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} g_\phi(v_j(t), \mathbf{a}_j)^2, I_i(t)\right) \quad (14)$$

where f_θ, g_ϕ are MLPs, $\hat{W}_{ij}$ are learnable edge weights, and $\mathbf{a}_i \in \mathbb{R}^2$ are learnable per-neuron embeddings. The student must simultaneously learn the dynamics *and* select a point from $[\mathbf{W}^*]$. Unlike E&C where the student architecture matches the teacher exactly, here the student approximates a continuous ODE with discrete MLP layers — a fundamental mismatch. Training minimizes:

$$\begin{aligned} \mathcal{L} = & \|\widehat{\mathbf{y}} - \mathbf{y}\|_2 + \lambda_0 \|\theta\|_1 + \lambda_1 \|\phi\|_1 + \lambda_2 \|\hat{\mathbf{W}}\|_1 \\ & + \gamma_0 \|\theta\|_2 + \gamma_1 \|\phi\|_2 \\ & + \mu_0 \|\text{ReLU}(-\partial_v g_\phi)\|_2 + \mu_1 \|g_\phi(v_\star, \mathbf{a}) - v_\star\|_2 \end{aligned} \quad (15)$$

The ℓ_1 penalty on $\hat{\mathbf{W}}$ acts as a regularizer that selects a sparse point from $[\mathbf{W}^*]$, analogously to E&C’s clustering step selecting the teacher neuron from the set of duplicates. The GNN’s message-passing inductive bias further constrains which member of $[\mathbf{W}^*]$ is found.

The gap between GNN ($R^2 = 0.899$, $\sigma = 0$) and pseudoinverse ($R^2 = 0.988$) quantifies the cost of *dynamic misspecification*, not the size of $[\mathbf{W}^*]$: the pseudoinverse has exact access to τ_i , V_i^{rest} , and $\dot{v}_i$; the GNN must approximate all three.

4.3 The noise level σ as an ease-of-identifiability axis

In E&C, the ratio r/d_{in} defines an ease-of-identifiability axis: low r/d_{in} means orthogonal teacher hyperplanes and easy recovery; high r/d_{in} means near-parallel hyperplanes, a rough loss landscape, and hard recovery. Here the noise level σ plays the identical role:

$$\sigma \uparrow \Rightarrow \text{rank}(\mathbf{H}_i) \uparrow \Rightarrow \dim[\mathbf{w}_i^*] \downarrow \Rightarrow R_{\mathbf{W}}^2 \uparrow \quad (16)$$

Noise	Activity rank r	$\dim[\mathbf{W}^*]$	% identifiable	GNN $R_{\mathbf{W}}^2$
$\sigma = 0$	45	115,223	73.5%	0.899 ± 0.030
$\sigma = 0.05$	134	10,227	97.6%	0.962 ± 0.012
$\sigma = 0.5$	781	0	100.0%	0.997 ± 0.000

Independent noise realizations break the duplicate-type symmetry: same-type neurons now receive distinct perturbations, so their activity columns in $\mathbf{H}_i$ become linearly independent, collapsing $[\mathbf{w}_i^*]$ to a singleton. At $\sigma = 0.5$, $\text{rank } 781 \gg d_i^{\text{max}} = 208$, so every neuron’s equivalence class is a singleton and $R_{\mathbf{W}}^2 = 0.997$.

Experimentally, the same collapse could be achieved by optogenetic injection of independent perturbations per neuron (e.g., distinct light patterns per L1 across columns), or selective ablation of same-type neurons across columns — both increase $\text{rank}(\mathbf{H}_i)$ and shrink $[\mathbf{w}_i^*]$ by the same mechanism as noise.

5 Summary

The equivalence class $[\mathbf{W}^*]$ of zero-rollout-loss connectomes is a continuous $\sim 121,100$ -dimensional affine manifold, arising from duplicate-type neuron groups in the columnar flyvis connectome ($\sim 28\%$ of 434,112 edges). This is the biological analogue of overparameterisation symmetry in teacher-student MLP recovery, but continuous rather than discrete. Two independent estimates

of $\dim[\mathbf{W}^*]$ — global SVD (115,223) and structural counting ($\sim 121,100$) — agree within 5%. Empirically, 1,960 variants sampled from $[\mathbf{W}^*]$ confirm the decoupling: connectivity R^2 ranges 0.28–1.0 while rollout $R^2 > 0.96$.

Multiple GNN seeds sample $[\mathbf{W}^*]$. Five independent training seeds converge to distinct members of $[\mathbf{W}^*]$, yielding connectivity $R^2 \in [0.934, 0.972]$ at fixed rollout quality. Their ensemble mean improves to $R^2 = 0.976$, as null-space components (random across seeds) cancel while the identifiable component is preserved. SVD of the inter-seed variation reveals a ≤ 4 -dimensional explored subspace of the 121,100-dimensional null space, with the dominant direction explaining 58% of inter-seed variance. Per-type seed SNR analysis (Table 2) identifies two mechanisms of low identifiability: null-space degeneracy (L1, L2; $\bar{k} \geq 4.5$, $\text{SNR} \leq 0.5$) and near-zero absolute weights (T1, Am, Lawf1; $\bar{k} = 1$ but $\text{SNR} < 0.7$). Types with strong synaptic weights recover consistently regardless of $\bar{k}$: R8 and CT1(Lo1) achieve $\text{SNR} > 90$.

Minimum-support extreme of $[\mathbf{W}^*]$. Collapsing each degenerate (i, α) group to a single calibrated representative zeros 308,160 edges (72.5%) while maintaining rollout Pearson $r = 0.991$. This is the minimum-support member of $[\mathbf{W}^*]$: the maximum-zeros point on the null-space manifold that still preserves dynamics. The 308,160 zeros correspond exactly to the structural null dimension of the 64,279 degenerate groups.

Recovery of $\mathbf{W}^*$ from $[\mathbf{W}^*]$ is nonetheless possible because the equivalence class is *approximate*: duplicate-type activities are correlated but not identical, and the pseudoinverse selects $\mathbf{W}^*$ with $R^2 = 0.988$ even at $\sigma = 0$. The noise level σ defines an ease-of-identifiability axis analogous to r/d_{in} in E&C, collapsing $[\mathbf{W}^*]$ monotonically: rank $45 \rightarrow 781$, GNN R^2 $0.899 \rightarrow 0.997$. The residual GNN gap from the pseudoinverse (0.988) quantifies dynamic misspecification, not the intrinsic size of $[\mathbf{W}^*]$.